

Math 290

Practice Quiz 2 — Version E Solutions

1(a) Answer: A. To prove $P \Rightarrow Q$ by contradiction, you assume P and $\neg Q$ — here $2 \mid x$ and $2 \nmid (x^2 + 1)$ — and reach a false statement (1 is even). The proof does exactly this, for an arbitrary x . (Based on Proposition 9.4 and Advice 9.3.)

1(b) Answer: True. Take $a = 8$, $b = 4$: $a^2 = 64$ divides $b^3 = 64$, but $8 \nmid 4$. (Based on Proposition 11.4.)

1(c) Answer: False. Counterexample $a = 2$, $b = -2$: $2 \mid -2$ and $-2 \mid 2$, but $2 \neq -2$. (Based on Section 11.D.)

2. (Based on Exercise 8.7.)

Solution. Let $a, b, c, d, n \in \mathbb{Z}$. Assume $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$. Then $a - b = nk$ and $c - d = n\ell$ for some $k, \ell \in \mathbb{Z}$. We find

$$ac - bd = ac - bc + bc - bd = c(a - b) + b(c - d) = cnk + bn\ell = n(ck + b\ell).$$

Since $ck + b\ell \in \mathbb{Z}$, we have $n \mid (ac - bd)$, so $ac \equiv bd \pmod{n}$.

Application: $19 - 5 = 14 = 7 \cdot 2$, so $19 \equiv 5 \pmod{7}$. Using the result with $a = c = 19$ and $b = d = 5$ gives $19^2 \equiv 5^2 \pmod{7}$. Using it again with $a = 19^2$, $b = 5^2$, $c = 19$, $d = 5$ gives $19^3 \equiv 5^3 \pmod{7}$. So **yes**. $\square$

3. (Based on Exercises 11.4 and 11.3 (with Proposition 8.1).)

Solution. (a) *Disproof.* We prove the negation: for every $x \in \mathbb{Z}$, $x^2 + x$ is even. Let $x \in \mathbb{Z}$. Case 1: $x = 2k$. Then $x^2 + x = 4k^2 + 2k = 2(2k^2 + k)$ is even. Case 2: $x = 2k + 1$. Then $x^2 + x = 4k^2 + 6k + 2 = 2(2k^2 + 3k + 1)$ is even. So no integer x makes $x^2 + x$ odd, and the statement is false.

(b) *Disproof.* Fix $s = 2$. Then $6s - 3 = 9$ is odd, but s is even. So the implication fails for $s = 2$, and the universal statement is false. (In fact $6s - 3 = 2(3s - 2) + 1$ is odd for every s .) $\square$

4. (Based on Sections 10.B and 11.D.)

Solution. (a) *Disproof.* Fix $A = \{1\}$, $B = \emptyset$, and $C = \{1\}$. Then $A \cup C = \{1\} \subseteq \{1\} = B \cup C$, but $1 \in A$ and $1 \notin B$, so $A \not\subseteq B$.

(b) Assume $A \cup C \subseteq B \cup C$ and $A \cap C \subseteq B \cap C$. Let $x \in A$. We know $x \in C$ or $x \notin C$ (law of the excluded middle).

Case 1: $x \in C$. Then $x \in A \cap C$ (definition of intersection). So $x \in B \cap C$ (definition of subset, since $A \cap C \subseteq B \cap C$). Hence $x \in B$ (definition of intersection).

Case 2: $x \notin C$. Since $x \in A$, we have $x \in A$ or $x \in C$ (tautology $P \Rightarrow P \vee Q$). So $x \in A \cup C$ (definition of union). So $x \in B \cup C$ (definition of subset, since $A \cup C \subseteq B \cup C$). Thus $x \in B$ or $x \in C$ (definition of union). Since $x \notin C$, we get $x \in B$ (disjunctive syllogism).

In both cases $x \in B$, so $A \subseteq B$ (definition of subset). $\square$